MAT2040

Tutorial 11

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Question 1

Let U and V be subspaces of a vector space W. Show that if $W = U \oplus V$, then $U \cap V = \{\mathbf{0}\}$.

Solution 1

Let $\mathbf{x} \in U \cap V$. We can write

$$\begin{aligned} \mathbf{x} &= \mathbf{0} + \mathbf{x} \quad (\mathbf{0} \in U, \quad \mathbf{x} \in V) \\ \mathbf{x} &= \mathbf{x} + \mathbf{0} \quad (\mathbf{x} \in U, \quad \mathbf{0} \in V) \end{aligned}$$

By the uniqueness of the direct sum representation $\mathbf{x} = \mathbf{0}$.

Question 2

Let $\mathbf{x}$ and $\mathbf{y}$ be linearly independent vectors in $\mathbb{R}^n$ and let $S = \text{Span}(\mathbf{x}, \mathbf{y})$. We can use x and y to define a matrix A by setting

$$A = \mathbf{x}\mathbf{y}^T + \mathbf{y}\mathbf{x}^T \tag{1}$$

(**a**) Show that A is symmetric.
(**b**) Show that $N(A) = S^{\perp}$.
(**c**) Show that the rank of A must be 2.

Solution 2

(a) A is symmetric since

$$A^T = (\mathbf{x}\mathbf{y}^T + \mathbf{y}\mathbf{x}^T)^T = (\mathbf{x}\mathbf{y}^T)^T + (\mathbf{y}\mathbf{x}^T)^T \tag{2}$$

$$= (\mathbf{y}^T)^T\mathbf{x}^T + (\mathbf{x}^T)^T\mathbf{y}^T = \mathbf{y}\mathbf{x}^T + \mathbf{x}\mathbf{y}^T = A \tag{3}$$

(b) For any vector z in R^n

$$A\mathbf{z} = \mathbf{x}\mathbf{y}^T\mathbf{z} + \mathbf{y}\mathbf{x}^T\mathbf{z} = c_1\mathbf{x} + c_2\mathbf{y} \tag{4}$$

where $c_1 = \mathbf{y}^T\mathbf{z}$ and $c_2 = \mathbf{x}^T\mathbf{z}$. If z is in $N(A)$ then

$$\mathbf{0} = A\mathbf{z} = c_1\mathbf{x} + c_2\mathbf{y} \tag{5}$$

and since $\mathbf{x}$ and y are linearly independent we have $\mathbf{y}^T\mathbf{z} = c_1 = 0$ and $\mathbf{x}^T\mathbf{z} = c_2 = 0$. So z is orthogonal to both x and y. Since x and y span S it follows that z$\in S^{\perp}$. Conversely, if z is in $S^{\perp}$ then z is orthogonal to both x and y. It follows that

$$A\mathbf{z} = c_1\mathbf{x} + c_2\mathbf{y} = 0 \tag{6}$$

since $c_1 = \mathbf{y}^T\mathbf{z} = 0$ and $c_2 = \mathbf{x}^T\mathbf{z} = 0$. Therefore $\mathbf{z}$ is in $N(A)$ and hence $N(A) = S^{\perp}$.

Solution 2

(c) Clearly dim $S = 2$ and by Theorem 5.2.2,dim S + dim $S^{\perp} = n$. Using our result from part (a) we have

$$\dim N(A) = \dim S^{\perp} = n - 2 \tag{7}$$

So A has nullity $n - 2$. It follows from the Rank-Nullity Theorem that the rank of A must be 2.

Question 3

Find the equation of the circle that gives the best least squares circle fit to the points (-1, -2),(0, 2.4),(1.1, -4) and (2.4, -1.6).

Solution 3

Assume that the circle equation is $(x_i - a)^2 + (y_i - b)^2 = r^2, i = 1, 2, 3, 4$, which can be rewritten as

$$2x_i a + 2y_i b + \left(r^2 - a^2 - b^2\right) = x_i^2 + y_i^2$$

Denote $r^2 - a^2 - b^2 \triangleq c$, and $\boldsymbol{x} = [a, b, c]^T$.

$$\boldsymbol{A} = \begin{bmatrix} 2x_1 & 2y_1 & 1 \\ 2x_2 & 2y_2 & 1 \\ 2x_3 & 2y_3 & 1 \\ 2x_4 & 2y_4 & 1 \end{bmatrix}, \boldsymbol{b} = \begin{bmatrix} x_1^2 + y_1^2 \\ x_2^2 + y_2^2 \\ x_3^2 + y_3^2 \\ x_4^2 + y_4^2 \end{bmatrix}$$

Solve $\hat{\boldsymbol{x}} = \left(\mathbf{A}^T\mathbf{A}\right)^{-1}\mathbf{A}^T\mathbf{b}$, we can get

$$(x - 0.5755)^2 + (y + 0.6426)^2 = 7.4265$$

Question 4

Let $A \in \mathbb{R}^{m \times n}$ and let $\hat{\mathbf{x}}$ be a solution of the least squares problem $A\mathbf{x} = \mathbf{b}$. Show that a vector $\mathbf{y} \in \mathbb{R}^n$ will also be a solution if and only if $\mathbf{y} = \hat{\mathbf{x}} + \mathbf{z}$, for some vector $\mathbf{z} \in N(A)$. [Hint: $N\left(A^T A\right) = N(A)$.]

Solution 4

If $\hat{\mathbf{x}}$ is a solution to the least squares problem, then $\hat{\mathbf{x}}$ is a solution to the normal equations

$$A^T A\mathbf{x} = A^T \mathbf{b}$$

It follows that a vector $\mathbf{y} \in R^n$ will be a solution if and only if

$$\mathbf{y} = \hat{\mathbf{x}} + \mathbf{z}$$

for some $\mathbf{z} \in N\left(A^T A\right)$. (See Exercise 20, Chapter 3, Section 6). Since

$$N\left(A^T A\right) = N(A)$$

we conclude that $\mathbf{y}$ is a least squares solution if and only if

$$\mathbf{y} = \hat{\mathbf{x}} + \mathbf{z}$$

for some $\mathbf{z} \in N(A)$.